

Earshot

Proximity, Not Transmission: What the Patent Record Can and Cannot Show About NLP and Surveillance.

TL;DR

We introduce Earshot, a purpose-based audit of patents citing language-technology research. We classify each patent as surveillance, military, dual-use, or neither, and assess what patent citations can support about downstream application.

Abstract

Researchers use patent citations to trace links between basic research and downstream technology, but citation alone cannot establish use or deployment. We introduce Earshot, a purpose-based audit of patents that cite language-technology research, to measure their proximity to surveillance and military applications. Earshot links 127,851 ACL Anthology papers to 9,048 USPTO patents and labels 2,374 randomly sampled patents as surveillance, military, dual-use, or neither. Its rubric separates operator from subject and authentication from identification, two distinctions that matter especially for speech systems. Two human coders independently label a stratified sample of 200 patents to evaluate classification reliability. Since 91.6% of citation links in our corpus appear only on the patent front page, compared with 62.3% in the wider citation record, we treat these links as evidence of proximity rather than transmission.

Earshot finds 11 surveillance patents among the 2,374 sampled patents, a rate of 0.46%. None of the ten subfields shows a concentration of surveillance labels, and we find no trend by year. Matching patents by grant era eliminates the apparent difference between patents citing language research and patents citing science more broadly. The uncertainty remains wide enough to include both no difference and a difference as large as threefold. Masking applicant names changes 10 of 31 non-neutral judgments, showing that assignee identity can alter assessments of patent purpose. We therefore find little observable surveillance in the patent record, but that result does not establish weak downstream involvement. Patent abstracts can omit decisive application context, and front-page citations cannot show whether inventors read or used the cited research. Our results show that citation counts can measure technical proximity, but they cannot establish what the citing patents are actually for.